

Year 12
Mathematics Extension 2
HSC Trial Examination
2015

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this paper
- In questions 11 – 16, show all relevant reasoning and/or calculations

Total marks – 100

Section I

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Section I

10 marks

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. Evaluate $\int \frac{dx}{x^2 - 4x + 13}$
- (A) $\frac{1}{3} \tan^{-1} \left(\frac{x-2}{3} \right) + C$
- (B) $\frac{2}{3} \tan^{-1} \left(\frac{x-2}{3} \right) + C$
- (C) $\frac{1}{3} \tan^{-1} \left(\frac{2x-4}{3} \right) + C$
- (D) $\frac{2}{3} \tan^{-1} \left(\frac{2x-4}{3} \right) + C$
2. The foci of the hyperbola $\frac{y^2}{8} - \frac{x^2}{12} = 1$ are:
- (A) $(\pm 2\sqrt{5}, 0)$
- (B) $(\pm \sqrt{30}, 0)$
- (C) $(0, \pm 2\sqrt{5})$
- (D) $(0, \pm \sqrt{30})$
3. The gradient of the curve $xy - x^2 + 3 = 0$ at the point when $x = 1$ is:
- (A) -4
- (B) -1
- (C) 1
- (D) 4

4. The region bounded by the curves $y = x^2$ and $y = x^3$ in the first quadrant is rotated about the y -axis. The volume of the solid of revolution formed can be found using:

(A) $V = \pi \int_0^1 \left(y^{\frac{1}{3}} - y^{\frac{1}{2}} \right) dy$

(B) $V = \pi \int_0^1 \left(y^{\frac{1}{2}} - y^{\frac{1}{3}} \right) dy$

(C) $V = \pi \int_0^1 \left(y^{\frac{2}{3}} - y \right) dy$

(D) $V = \pi \int_0^1 (x^4 - x^6) dx$

5. The five fifth roots of $1 + \sqrt{3}i$ are:

(A) $2^{\frac{1}{5}} \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{15} \right), k = 0, 1, 2, 3, 4$

(B) $2^5 \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{15} \right), k = 0, 1, 2, 3, 4$

(C) $2^{\frac{1}{5}} \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{30} \right), k = 0, 1, 2, 3, 4$

(D) $2^5 \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{30} \right), k = 0, 1, 2, 3, 4$

6. The locus of a complex number z is the line $4x - 3y - 12 = 0$

What is the minimum value of $|z|$?

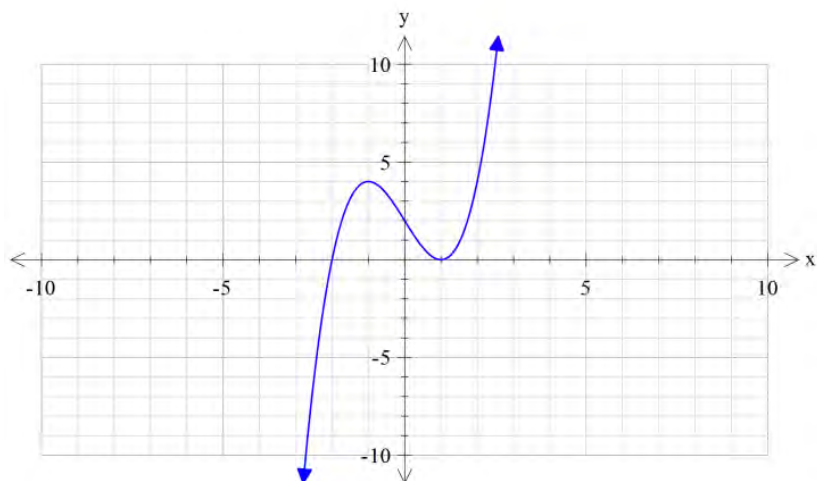
(A) $\frac{12}{5}$

(B) 3

(C) 4

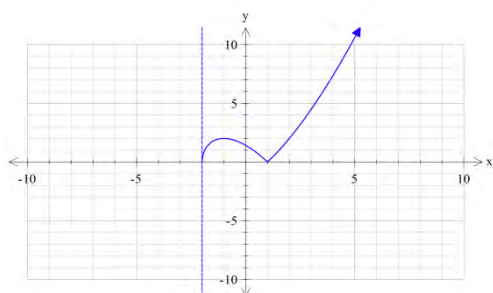
(D) 5

7. The diagram of $y = f(x)$ is drawn below.

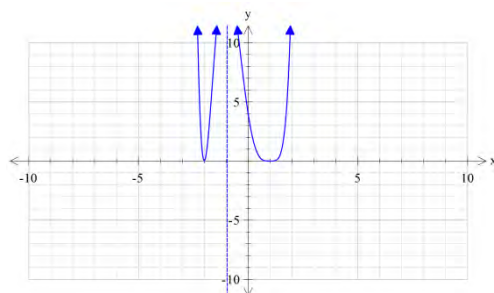


Which of the diagrams below best represents $y = \sqrt{f(x)}$

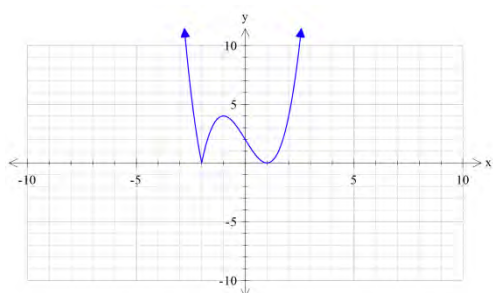
(A)



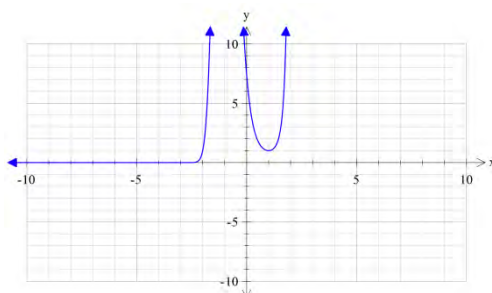
(B)



(C)



(D)



8. An object of mass 5kg is tied to a piece of rope, 3 metres in length, which has a breaking strain of 240N.
The rope is then swung in a horizontal circle.
What is the angular velocity of the object at the moment the rope breaks?
- (A) 2
(B) 4
(C) 8
(D) 16
9. What is the remainder when $P(x) = x^3 + x^2 - x + 1$ is divided by $(x - 1 - i)$?
- (A) $-3i - 2$
(B) $3i - 2$
(C) $3i + 2$
(D) $2 - 3i$
10. Solve the inequality: $\frac{x+1}{x-3} \leq \frac{x+3}{x-2}$.
- (A) $x < 2$ and $x > 3$
(B) $x < 2$ and $3 < x \leq 7$
(C) $2 < x < 3$
(D) $2 < x < 3$ and $x \geq 7$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $A = 3 + 3\sqrt{3}i$ and $B = -5 - 12i$. Find the value of:

(i) $\bar{B}$ **1**

(ii) $\frac{A}{B}$ **2**

(iii) $\sqrt{B}$ **2**

(iv) The modulus and argument of A **2**

(v) A^4 **1**

(b) The roots of the polynomial equation $2x^3 - 3x^2 + 4x - 5 = 0$ are α , β and γ . Find the polynomial equation which has roots:

(i) $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$. **2**

(ii) 2α , 2β and 2γ . **2**

(c) Find $\int \frac{dx}{\sqrt{9 + 16x - 4x^2}}$. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^{\frac{\sqrt{\pi}}{2}} 3x \sin(x^2) dx$. **3**

(b) (i) Find the values of A , B , and C such that: **2**

$$\frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+2}$$

(ii) Hence evaluate $\int \frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} . dx$ **2**

(c) Solve the equation $x^4 - 7x^3 + 17x^2 - x - 26 = 0$, given that $x = (3 - 2i)$ is a root of the equation. **3**

(d) (i) Show that the equation of the tangent at the point $P\left(ct, \frac{c}{t}\right)$ on the rectangular hyperbola $xy = c^2$ is $x + t^2y - 2ct = 0$. **2**

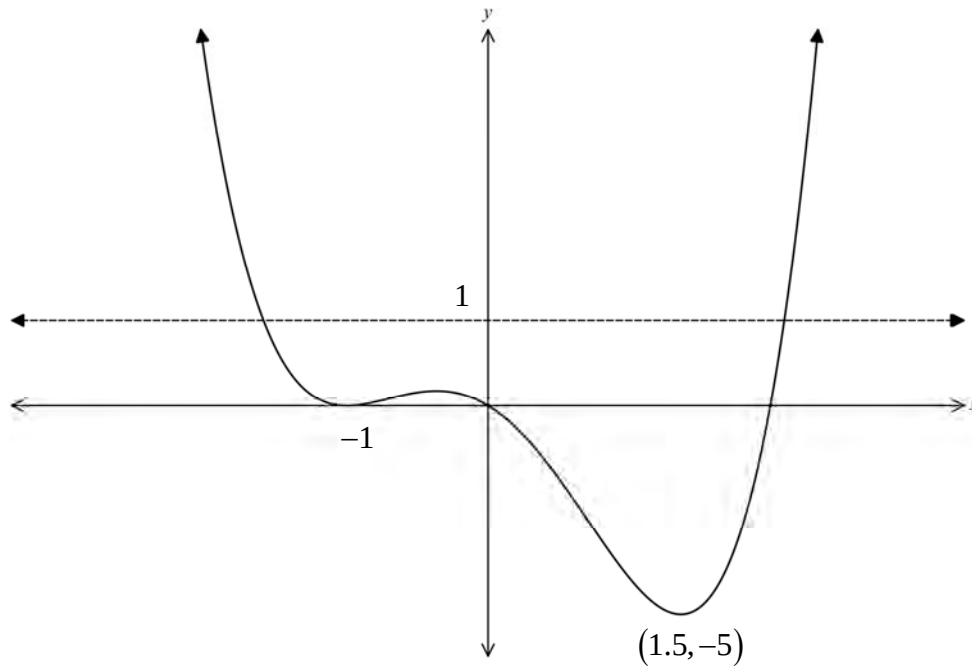
(ii) Find the coordinates of A and B where this tangent cuts the x and y axes respectively. **2**

(iii) Prove that the area of the triangle OAB is a constant, where O is the origin. **1**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) The graph of $y = f(x)$ is shown below.



Draw separate sketches for each of the following:

(i) $y = |f(x)|$ **1**

(ii) $y = \frac{1}{f(x)}$ **2**

(iii) $y^2 = f(x)$ **2**

(iv) $y = e^{f(x)}$ **2**

(b) Show that the equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point **3**

$P(x_1, y_1)$ is given by the equation: $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$.

Question 13 continues on the next page.

- (c) A particle of unit mass is projected vertically upwards. The resistance to the motion is proportional to the square of the velocity. The velocity of projection is V .

(i) Show that the acceleration is given by: $\ddot{x} = -(g + kv^2)$. **1**

(ii) Show that the maximum height H reached is: **2**

$$H = \frac{1}{2k} \ln \left\{ \frac{(g + kV^2)}{(g)} \right\}$$

(iii) Show that T , the time taken to reach H is: **2**

$$T = \frac{1}{\sqrt{kg}} \tan^{-1} \left(\frac{\sqrt{k}}{\sqrt{g}} \right) V$$

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Show that: $\frac{\cos A - \cos(A+2B)}{2 \sin B} = \sin(A + B)$. **3**
- (b) A mass of 5kg, on the end of a string 0.5 metre long, is rotating in a conical pendulum with angular velocity 2π radians per second. Use $g = 10 \text{ m/s}^2$ and let θ be the angle that the string makes with the vertical.
- (i) Draw a diagram showing all the forces acting on the mass. **1**
- (ii) By resolving forces, find the tension in the string. **2**
- (iii) Find θ , correct to the nearest degree. **1**
- (c) A sequence is defined such that $u_1 = 1, u_2 = 1$ and $u_n = u_{n-1} + u_{n-2}$ for $n \geq 3$. **4**
- Prove by induction that $u_n < \left(\frac{7}{4}\right)^n$ for integers $n \geq 1$.
- (d) Use the method of cylindrical shells to find the volume of the solid generated by revolving the region enclosed by $y = 3x^2 - x^3$ and the x axis around the y-axis. **4**

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Derive the reduction formula:

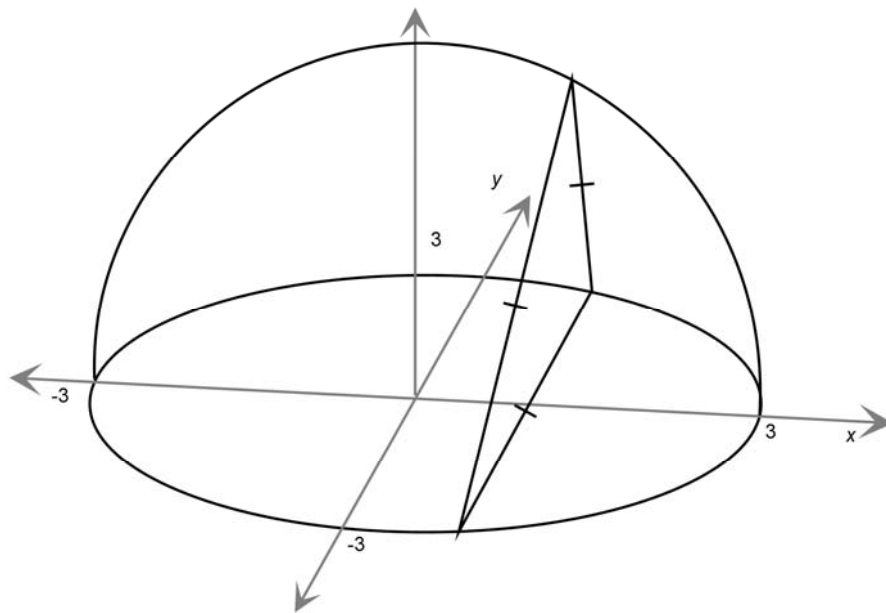
2

$$\int x^n e^{-x^2} dx = -\frac{1}{2} x^{n-1} e^{-x^2} + \frac{n-1}{2} \int x^{n-2} e^{-x^2} dx$$

- (ii) Use this reduction formula to evaluate $\int_0^1 x^5 e^{-x^2} dx$

2

- (b)



The diagram above shows a solid which has the circle $x^2 + y^2 = 9$ as its base.

The cross-section perpendicular to the x axis is an equilateral triangle.

- (i) Show that the area of a triangle is given by:

2

$$\text{Area} = \sqrt{3} (9 - x^2)$$

- (ii) Hence or otherwise find the volume of the solid.

2

Question 15 continues on the next page.

(c) Given that $x^4 - 6x^3 + 9x^2 + 4x - 12 = 0$, has a double root at $x = \alpha$, find the value of α . **3**

(d) If z represents the complex number $x + iy$, sketch the regions:

(i) $|\arg z| < \frac{\pi}{4}$ **2**

(ii) $\operatorname{Im}(z^2) = 4$ **2**

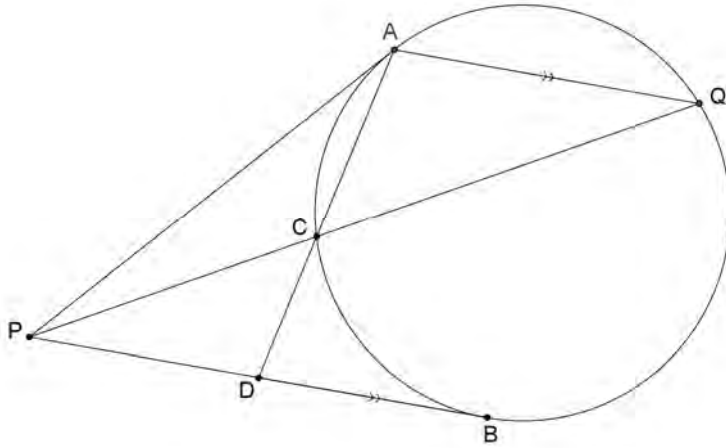
End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the hyperbola with equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a > b$.
- (i) Show that the equation of the tangent at the point $P(a \sec \theta, b \tan \theta)$ has the equation $bx \sec \theta - ay \tan \theta = ab$. **2**
- (ii) Find the equation of the normal at P . **2**
- (iii) Find the coordinates of the points A and B where the tangent and normal respectively cut the y -axis. **2**
- (iv) Show that AB is the diameter of the circle that passes through the foci of the hyperbola. **3**
- (b) Five letters are chosen from the letters of the word *CHRISTMAS*. **2**
- These five letters are then placed alongside one another to form a five letter arrangement.
- Find the number of distinct five letter arrangements which are possible, considering all choices.

Question 16 continues on the next page.

- (c) In the diagram below, PA and PB are tangents to the circle. The chord AQ is parallel to the tangent PB . PCQ is a secant to the circle and chord AC produced meets PB at D .



- i) Show that $\triangle CDP$ is similar to $\triangle PDA$. **2**

- ii) Show that $PD^2 = AD \times CD$ and hence, or otherwise, prove that AD bisects PB . **2**

End of Examination.

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

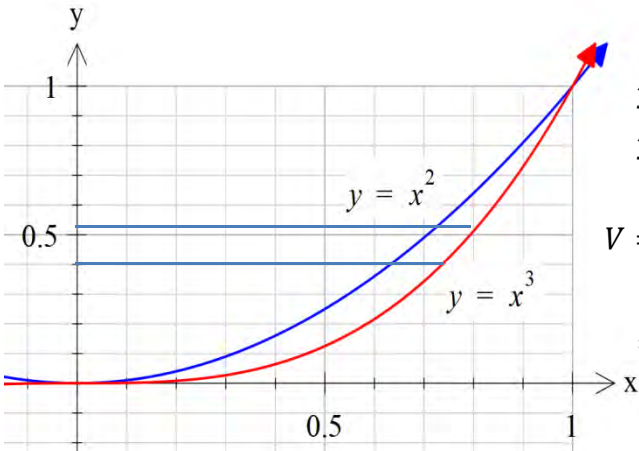
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2015 Extension 2 Trial solutions

Multiple Choice Worked Solutions

No	Working	Answer
1	$\int \frac{dx}{x^2-4x+13} = \int \frac{dx}{x^2-4x+4+9}$ $= \int \frac{dx}{(x-2)^2+9}$ $= \frac{1}{3} \tan^{-1} \left(\frac{x-2}{3} \right) + C$	A
2	$\frac{y^2}{8} - \frac{x^2}{12} = 1$ $a = 2\sqrt{2}, b = 2\sqrt{3}$ $b^2 = a^2(e^2 - 1)$ $(2\sqrt{3})^2 = (2\sqrt{2})^2(e^2 - 1)$ $12 = 8(e^2 - 1)$ $\frac{12}{8} = e^2 - 1$ $e^2 = \frac{20}{8} = \frac{5}{2}$ $e = \frac{\sqrt{10}}{2}$ $\text{Foci} = (0, \pm ae) = \left(0, \pm 2\sqrt{2} \left(\frac{\sqrt{10}}{2} \right) \right) = (0, \pm \sqrt{20}) = (0, \pm 2\sqrt{5})$	C
3	$xy - x^2 + 3 = 0 \quad \text{when } x = 1, y - 1 + 3 = 0$ $x \frac{dy}{dx} + y - 2x = 0 \quad y = -2$ $x \frac{dy}{dx} = 2x - y$ $\frac{dy}{dx} = \frac{2x-y}{x} \quad \text{At } (1, -2) \quad \frac{dy}{dx} = \frac{2(1) - (-2)}{1} = 2 + 2 = 4$	D
4	 $y = x^3 \rightarrow x = y^{\frac{1}{3}}$ $y = x^2 \rightarrow x = y^{\frac{1}{2}}$ $V = \pi \int_0^1 \left[\left(y^{\frac{1}{3}} \right)^2 - \left(y^{\frac{1}{2}} \right)^2 \right] dy$ $= \pi \int_0^1 \left(y^{\frac{2}{3}} - y \right) dy$	C
5	$z^5 = 1 + \sqrt{3}i$ $R = \sqrt{1^2 + (\sqrt{3})^2} = 2$ $\text{Arg } z: \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ $\therefore z^5 = 2 \text{ cis } \frac{\pi}{3}$ $z = 2^{\frac{1}{5}} \text{ cis } \left(\frac{2k\pi}{5} + \frac{\pi}{15} \right), k = 0, 1, 2, 3, 4$	A

6	z represents the length of the vector from the origin to z. Hence the minimum distance from the origin to z is the perpendicular distance from $(0, 0)$ to $4x - 3y - 12 = 0$ $d = \left \frac{0 + 0 - 12}{\sqrt{4^2 + (-3)^2}} \right = \left \frac{12}{5} \right = \frac{12}{5}$	A
7	Graph A	A
8	$F = mr\omega^2$ $240 = 5 \times 3 \times \omega^2$ $240 = 15\omega^2$ $16 = \omega^2$ $\omega = 4$	B
9	$P(x) = x^3 + x^2 - x + 1$ is divided by $(x - 1 - i)$ Let $x = 1 + i$ $x^2 = (1 + i)^2 = 1 + 2i + i^2 = 2i$ $x^3 = 2i(1 + i) = 2i + 2i^2 = 2i - 2$ Remainder = $P(1 + i) = 2i - 2 + 2i - (1 + i) + 1$ $= 4i - 1 - 1 - i$ $= 3i - 2$	B
10	$\frac{x+1}{x-3} \leq \frac{x+3}{x-2}$ $x \neq 3$ or 2 Then $(x+1)(x-2) = (x+3)(x-3)$ $x^2 - 2x + x - 2 = x^2 - 9$ $x^2 - x - 2 = x^2 - 9$ $-x = -7$ $x = 7$ By inspection, $2 < x < 3 \cap x \geq 7$	D

- | | | | | | | | | |
|-----|---|----------------------------------|---|----------------------------------|---|----------------------------------|---|----------------------------------|
| 1. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 4. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 5. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 7. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |

Question 11		2015	
	Solution	Marks	Allocation of marks
(a)	$A = 3 + 3\sqrt{3}i$ and $B = -5 - 12i$. (i) $\bar{B} = \overline{-5 - 12i}$ $= -5 + 12i$ (ii) $\frac{A}{B} = \frac{3+3\sqrt{3}i}{-5-12i}$ $\frac{A}{B} = \frac{3+3\sqrt{3}i}{-5-12i} \times \frac{-5+12i}{-5+12i}$ $= \frac{-15 + 36i - 15\sqrt{3}i - 36\sqrt{3}}{25 - 144i^2}$ $= \frac{(-15 - 36\sqrt{3}) + (36 - 15\sqrt{3})i}{169}$ (iii) $\sqrt{B} = \sqrt{-5 - 12i}$ Let $(x + iy)^2 = -5 - 12i$ $\therefore x^2 + 2ixy - y^2 = -5 - 12i$ $\therefore x^2 - y^2 = -5$ ----- (1) and $2xy = -12$ $(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4ixy$ $= (-5)^2 + (-12)^2$ $= 169$ $\therefore x^2 + y^2 = 13$ ----- (2) (1) + (2) $2x^2 = 8 \rightarrow x^2 = 4 \rightarrow x = \pm 2$ (2) - (1) $2y^2 = 18 \rightarrow y^2 = 9 \rightarrow y = \pm 3$ Since $2xy = -12$ $\sqrt{B} = \sqrt{-5 - 12i} = \pm(2 - 3i)$ (iv) Modulus $(r) = \sqrt{(3)^2 + (3\sqrt{3})^2} = \sqrt{36} = 6$ Argument: $\tan \theta = \frac{3\sqrt{3}}{3} = \sqrt{3}, \theta = \frac{\pi}{3}$ (v) $A^4 = \left(6 \operatorname{cis} \frac{\pi}{3}\right)^4 = 1296 \operatorname{cis} \frac{4\pi}{3} = 1296 \operatorname{cis} \frac{-2\pi}{3}$	1 2 2 2 1	Answer 1 – correct product 1 – correct answer 1 – working 1 – Answer 1 - modulus 1 - argument Correct answer

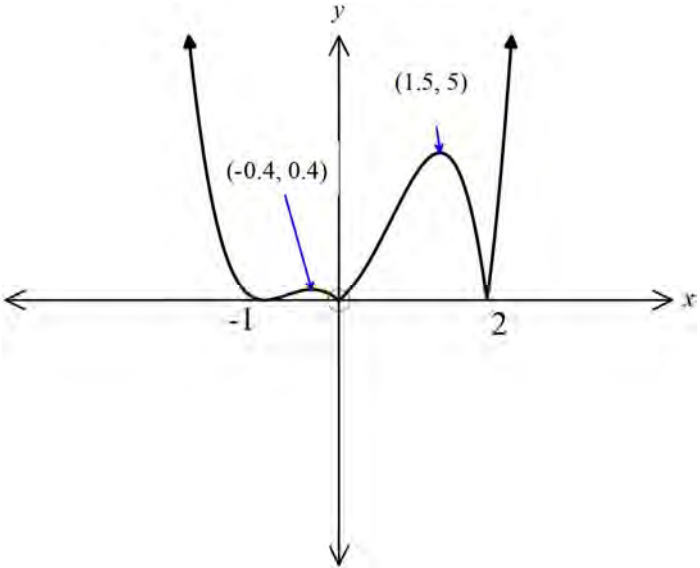
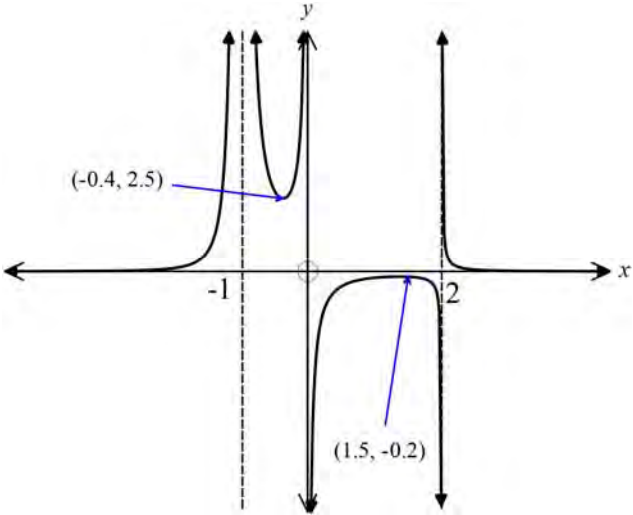
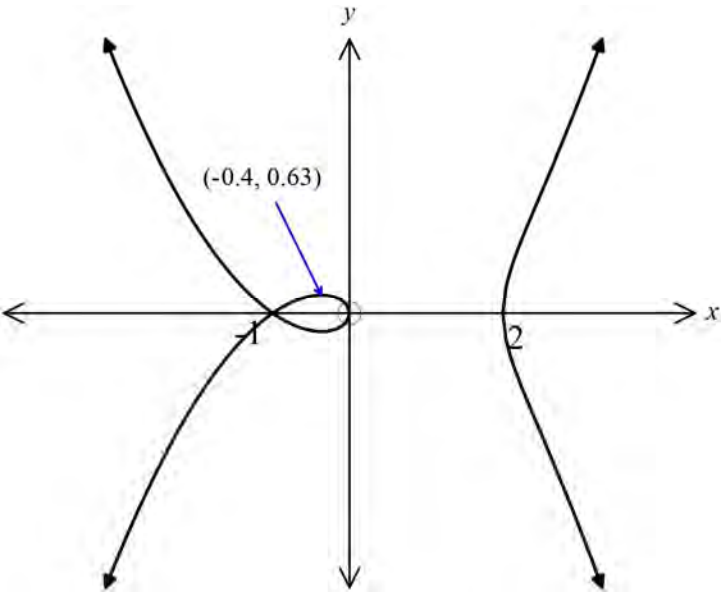
Question 11		2015	
	Solution	Marks	Allocation of marks
(b)	(i) $2x^3 - 3x^2 + 4x - 5 = 0$ Let $X = \frac{1}{x}$, $\therefore x = \frac{1}{X}$ Therefore equation is $2\left(\frac{1}{X}\right)^3 - 3\left(\frac{1}{X}\right)^2 + 4\left(\frac{1}{X}\right) - 5 = 0$ i.e. $\frac{2}{X^3} - \frac{3}{X^2} + \frac{4}{X} - 5 = 0$ Multiply by X^3 $2 - 3X + 4X^2 - 5X^3 = 0$ ie $5x^3 - 4x^2 + 3x - 2 = 0$ (ii) $2x^3 - 3x^2 + 4x - 5 = 0$ Let $X = 2x$ $\therefore x = \frac{X}{2}$ Therefore equation is $2\left(\frac{X}{2}\right)^3 - 3\left(\frac{X}{2}\right)^2 + 4\left(\frac{X}{2}\right) - 5 = 0$ $2\left(\frac{X^3}{8}\right) - 3\left(\frac{X^2}{4}\right) + \frac{4X}{2} - 5 = 0$ $\frac{X^3}{4} - \frac{3X^2}{4} + 2X - 5 = 0$ $X^3 - 3X^2 + 8X - 20 = 0$ i.e. $x^3 - 3x^2 + 8x - 20 = 0$	2 2	1 – correct substitution 1 – correct equation 1 – correct substitution 1 – correct equation

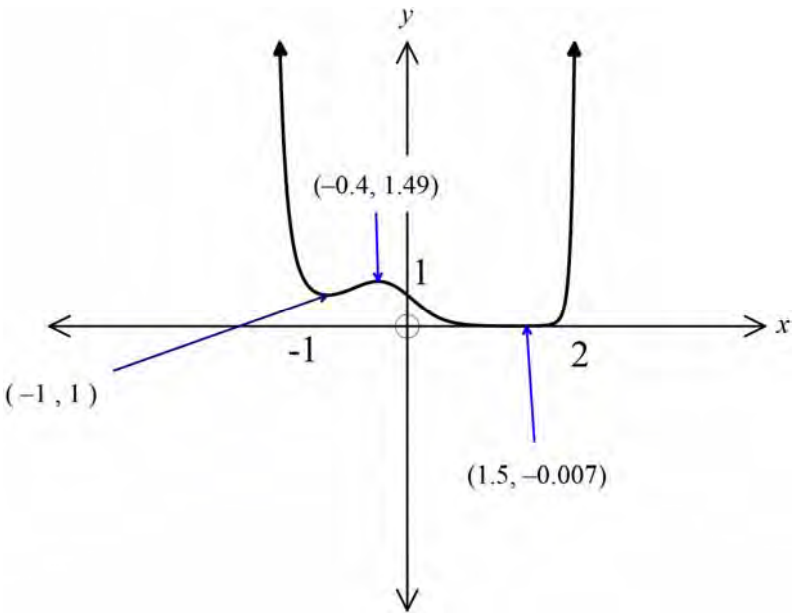
Question 11		2015	
	Solution	Marks	Allocation of marks
(c)	$\int \frac{dx}{\sqrt{9 + 16x - 4x^2}}$ $9 + 16x - 4x^2 = 9 - 4(x^2 - 4x)$ $= 9 - 4(x^2 - 4x + 4) + 16$ $= 25 - 4(x - 2)^2$ $\int \frac{dx}{\sqrt{9 + 16x - 4x^2}} = \int \frac{dx}{\sqrt{25 - 4(x-2)^2}}$ $= \frac{1}{5} \int \frac{dx}{\sqrt{1 - \frac{4}{25}(x-2)^2}}$ $u = \frac{2(x-2)}{5} + c$ $du = \frac{2}{5} dx$ $dx = \frac{5}{2} du$ $= \frac{1}{5} \int \frac{\frac{5}{2} du}{\sqrt{1 - u^2}}$ $= \frac{1}{2} \int \frac{du}{\sqrt{1 - u^2}}$ $= \frac{1}{2} \sin^{-1} u$ $= \frac{1}{2} \sin^{-1} \left(\frac{2(x-2)}{5} \right)$	3	1 – correct manipulation 1 – correct substitution 1 – correct answer

Question 12		2015	
	Solution	Marks	Allocation of marks

Question 12		2015	
	Solution	Marks	Allocation of marks
(b)	(i) $\frac{4x^2-3x-4}{x^3+x^2-2x} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+2}$ $\therefore 4x^2 - 3x - 4 = A(x-1)(x+2) + Bx(x+2) + Cx(x-1)$ When $x = 0$, $-4 = -2A \quad \therefore A = 2$ $x = -2$, $18 = 6C \quad \therefore C = 3$ $x = 1$, $-3 = 3B \quad \therefore B = -1$ $\therefore \frac{4x^2-3x-4}{x^3+x^2-2x} = \frac{2}{x} - \frac{1}{x-1} + \frac{3}{x+2}$ (ii) $\int \frac{4x^2-3x-4}{x^3+x^2-2x} = \int \left(\frac{2}{x} - \frac{1}{x-1} + \frac{3}{x+2} \right) dx$ $= 2\ln x - \ln(x-1) + 3\ln(x+2) + c$	2	1 - Working 1 – correct values 1 – correct integral 1 – correct answer
(c)	$x^4 - 7x^3 + 17x^2 - x - 26 = 0$ $(3 - 2i)$ is a factor $\therefore (3 + 2i)$ is also a factor since coefficients are real $\therefore x^2 - 6x + 13$ is a factor. By division, $x^4 - 7x^3 + 17x^2 - x - 26 = (x^2 - 6x + 13)(x^2 - x - 2)$ $= (x^2 - 6x + 13)(x - 2)(x + 1)$ Therefore solution to $x^4 - 7x^3 + 17x^2 - x - 26 = 0$ is: $x = 3 \pm 2i, -1 \text{ and } 2$ OR USE SUMS AND PRODUCTS OF ROOTS $\alpha = 3 - 2i, \beta = 3 + 2i, \gamma = ?, \delta = ?$ $\sum \alpha = 6 + \gamma + \delta \rightarrow \gamma + \delta = 1$ $\prod \alpha = 13\gamma\delta = -26 \rightarrow \gamma\delta = -2$ $\delta = -\frac{2}{\gamma}$ so $\gamma - \frac{2}{\gamma} = 1$ $\gamma^2 - \gamma - 2 = 0$ $\gamma = 2, -1$ $\therefore$ roots are $3 - 2i, 3 + 2i, 2, -1$	3	1 – using conjugate theorem Method 1 1 – division 1 – answer Method 2 1 correct use of sums and products 1 answer

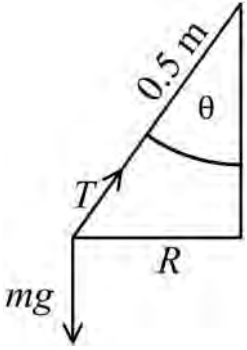
Question 12		2015	
	Solution	Marks	Allocation of marks
(d)	(i) $xy = c^2$ $P \left(ct, \frac{c}{t} \right)$ By implicit differentiation $y + x \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{y}{x}$ At $P \left(ct, \frac{c}{t} \right)$ $\frac{dy}{dx} = -\frac{c}{t} \div ct$ $= -\frac{1}{t^2}$ $y - y_1 = m(x - x_1)$ $y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$ $t^2 y - ct = -x + ct$ $x + t^2 y - 2ct = 0$	2	1 – gradient of tangent 1 – equation of tangent
	(ii) When $y = 0$, $x + 0 - 2ct = 0$ $x = 2ct$ $\therefore A(2ct, 0)$ When $x = 0$, $0 + t^2 y - 2ct = 0$ $y = \frac{2ct}{t^2} = \frac{2c}{t}$ $\therefore B \left(0, \frac{2c}{t} \right)$	2	One mark for each coordinate
	(iii) Now $OA = 2ct$ $OB = \frac{2c}{t}$ Area Triangle OAB $= \frac{1}{2} (2ct) \left(\frac{2c}{t} \right)$ $= 2c^2 \text{ which is a constant as } c \text{ is a constant.}$	1	Correct area

Question 13		2014	
	Solution	Marks	Allocation of marks
(a)	(i) 	1	1 – correct graph all coords shown
	(ii) 	2	1 – vertical asymptotes 1 – correct graph all coords shown
	(iii) 	2	1 – correct shape one side of axes 1 - correct graph both sides

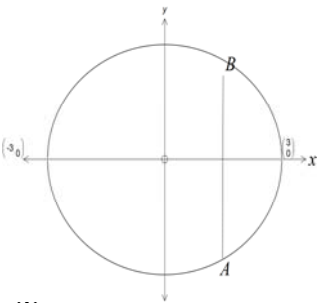
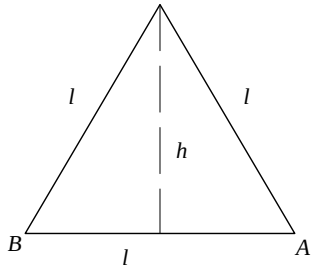
Question 13		2014	
	Solution	Marks	Allocation of marks
	(iv) 	2	1 – correct behaviour $x \rightarrow \infty$ 1 - correct graph all coords shown
(b)	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{-2b^2x}{2a^2y}$ At $P(x_1, y_1)$ $\frac{dy}{dx} = \frac{-b^2x_1}{a^2y_1}$ Normal $m = \frac{a^2y_1}{b^2x_1}$ $y - y_1 = m(x - x_1)$ $y - y_1 = \frac{a^2y_1}{b^2x_1}(x - x_1)$ $b^2x_1y - b^2x_1y_1 = a^2y_1x - a^2y_1x_1$ $a^2y_1x - b^2x_1y = a^2y_1x_1 - b^2x_1y_1$ ($\div x_1y_1$) $\therefore \frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$	3	1 – gradient of normal 1 use of equation 1 – completion of proof
(c)	(i) $m\ddot{x} = -mg - mkv^2$ $\ddot{x} = -g - kv^2$ $\ddot{x} = -(g + kv^2)$	1	Correct answer

Question 13		2014	
	Solution	Marks	Allocation of marks
	(ii) $v \frac{dv}{dx} = -(g + kv^2)$ $\frac{dv}{dx} = \frac{-(g+kv^2)}{v}$ $\frac{dx}{dv} = -\frac{v}{(g + kv^2)}$ $x = \int -\frac{v}{(g + kv^2)} dv$ $x = -\frac{1}{2k} \int \frac{2kv}{(g + kv^2)} dv$ $x = -\frac{1}{2k} \ln(g + kv^2)$ at $x = 0, v = V$ $0 = -\frac{1}{2k} \ln(g + kV^2) + C$ $\therefore C = \frac{1}{2k} \ln(g + kV^2)$ $x = -\frac{1}{2k} \ln(g + kv^2) + \frac{1}{2k} \ln(g + kV^2)$ $x = \frac{1}{2k} \ln \left\{ \frac{(g + kV^2)}{(g + kv^2)} \right\}$ Maximum height is obtained when $v = 0$. $H = \frac{1}{2k} \ln \left\{ \frac{(g + kV^2)}{g} \right\}$ (iii) $\frac{dv}{dt} = -(g + kv^2)$ $\frac{dt}{dv} = \frac{-1}{(g + kv^2)}$ $t = \int \frac{-1}{(g + kv^2)} dv$ $= -\frac{1}{k} \int \frac{1}{\frac{g}{k} + v^2} dv$ $t = -\frac{1}{k} \times \frac{1}{\sqrt{\frac{g}{k}}} \tan^{-1} \left(\frac{v}{\sqrt{\frac{g}{k}}} \right) + C_1$ A $t = -\frac{1}{k} \times \frac{\sqrt{k}}{\sqrt{g}} \tan^{-1} \left(\frac{\sqrt{k}v}{\sqrt{g}} \right) + C_1$ $A = -\frac{1}{k} \times \frac{\sqrt{k}}{\sqrt{g}} \tan^{-1} \left(\frac{\sqrt{k}v}{g} \right) + C_1$	4	1 – Evaluating integral 1 expression for H 1 evaluating integral

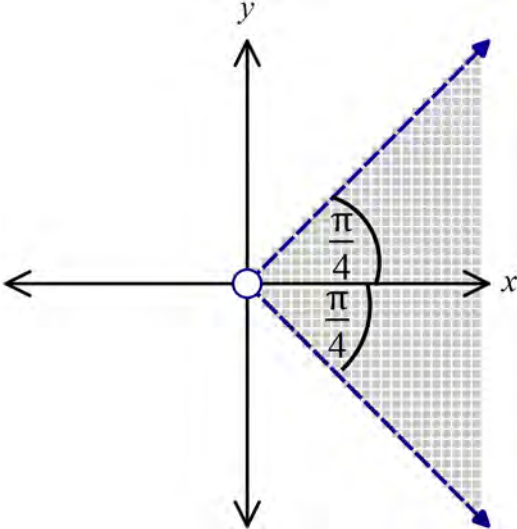
Question 13		2014	
	Solution	Marks	Allocation of marks
	$At\ t = 0, v = V$ $0 = -\frac{1}{\sqrt{kg}} \tan^{-1}\left(\frac{\sqrt{k}}{\sqrt{g}}\right) V + C_1$ $C_1 = \frac{1}{\sqrt{kg}} \tan^{-1}\left(\frac{\sqrt{k}}{\sqrt{g}}\right) V$ $t = -\frac{1}{\sqrt{kg}} \tan^{-1}\left(\frac{\sqrt{k}}{\sqrt{g}}\right) v + \frac{1}{\sqrt{kg}} \tan^{-1}\left(\frac{\sqrt{k}}{\sqrt{g}}\right) V$ Maximum height reached when $v = 0$, i.e. $T = \frac{1}{\sqrt{kg}} \tan^{-1}\left(\frac{\sqrt{k}}{\sqrt{g}}\right) V$		1 expression for T

Question 14		2014	
	Solution	Marks	Allocation of marks
(a)	$\frac{\cos A - \cos(A + 2B)}{2 \sin B} = \sin(A + B)$ $LHS = \frac{\cos A - \cos(A + 2B)}{2 \sin B}$ $= \frac{\cos A - (\cos A \cos 2B - \sin A \sin 2B)}{2 \sin B}$ $= \frac{\cos A - \cos A (1 - 2\sin^2 B) + 2 \sin A \sin B \cos B}{2 \sin B}$ $= \frac{\cos A - \cos A + 2\sin^2 B \cos A + 2 \sin A \sin B \cos B}{2 \sin B}$ $= \frac{2\sin^2 B \cos A + 2 \sin A \sin B \cos B}{2 \sin B}$ $= \frac{2 \sin B (\sin B \cos A + \sin A \cos B)}{2 \sin B}$ $= \sin B \cos A + \sin A \cos B$ $= \sin A \cos B + \sin B \cos A$ $= \sin(A + B)$ $= RHS$ $\therefore \frac{\cos A - \cos(A + 2B)}{2 \sin B} = \sin(A + B)$	3	1 -Using cosine double angle 1 – working 1 – completion of proof
(b)	(i)  $\sin \alpha = \frac{R}{0.5}$ $R = 0.5 \sin \alpha$	1	diagram
	(ii) $T \sin \theta = 5 \times (2\pi)^2 \times 0.5 \sin \theta$ $T = 5 \times 4\pi^2 \times 0.5$ $= 98.696$ $= 99\text{N (nearest newton) OR } 10\pi^2\text{N}$	2	1 – substitution 1 - answer
	(iv) $T \cos \theta = mg$ $= (5)(10)$ $= 50$ $\cos \theta = \frac{50}{T}$ $= \frac{50}{98.696}$ $\theta = 59.562$ $= 60^\circ \text{ (nearest degree)}$	1	Correct working to answer

Question 14		2014	
	Solution	Marks	Allocation of marks
(c)	Step1 Prove true for $n = 1$ and $n = 2$ $u_1 = 1 < \left(\frac{7}{4}\right)^1$ $\therefore$ true for $n = 1$ $u_2 = 1 < \left(\frac{7}{4}\right)^2$ $\therefore$ true for $n = 2$ Step 2 Let $n = k$ and $n = k - 1$ be values for which the statement is true ie $u_k < \left(\frac{7}{4}\right)^k$ and $u_{k-1} < \left(\frac{7}{4}\right)^{k-1}$ Step 3 Prove true for $n = k + 1$ $ie u_{k+1} < \left(\frac{7}{4}\right)^{k+1}$ $u_{k+1} = u_k + u_{k-1}$ $\therefore u_{k+1} < \left(\frac{7}{4}\right)^k + \left(\frac{7}{4}\right)^{k-1} \text{ from Step 2}$ $u_{k+1} < \left(\frac{7}{4}\right)^{k-1} \left(\frac{7}{4} + 1\right)$ $u_{k+1} < \left(\frac{7}{4}\right)^{k-1} \left(\frac{11}{4}\right)$ now $1 < \frac{11}{4} < \left(\frac{7}{4}\right)^2$ so the value of the RHS has increased $\therefore u_{k+1} < \left(\frac{7}{4}\right)^{k-1} \left(\frac{49}{16}\right)$ $u_{k+1} < \left(\frac{7}{4}\right)^{k-1} \left(\frac{7}{4}\right)^2$ $u_{k+1} < \left(\frac{7}{4}\right)^{k+1} \text{ as required}$ $\therefore$ true by mathematical induction	4	1 – $n = k, n = k - 1$ or similar having tested for $n = 1$ and $n = 2$ 1 $n = k + 1$ or similar (following on logically from Step 2) 1 – using assumption in Step 2 1 – proving true for $n = k + 1$ including justifying truth of inequality

Question 15		2014	
	Solution	Marks	Allocation of marks
(a)	$\int x^n e^{-x^2} dx$ (i) Let $u = x^{n-1}$ $v' = x e^{-x^2}$ $u' = (n-1)x^{n-2}$ $v = -\frac{1}{2} e^{-x^2}$ $\int x^n e^{-x^2} dx = uv - \int vu'$ $= -\frac{1}{2} x^{n-1} e^{-x^2} + \frac{n-1}{2} \int x^{n-2} e^{-x^2} dx$ (ii) $\int_0^1 x^5 e^{-x^2} dx = \left[-\frac{1}{2} x^4 e^{-x^2} \right]_0^1 + \frac{4}{2} \int_0^1 x^3 e^{-x^2} dx$ $= \frac{-1}{2e} + 2 \int_0^1 x^3 e^{-x^2} dx$ $= \frac{-1}{2e} + 2 \left\{ \left[-\frac{1}{2} x^2 e^{-x^2} \right]_0^1 + 1 \int_0^1 x e^{-x^2} dx \right\}$ $= \frac{-1}{2e} - 2 \left(\frac{1}{2e} \right) + 2 \left[-\frac{1}{2} x^0 e^{-x^2} \right]_0^1 + \frac{1-1}{2} \int x^{1-2} e^{-x^2} dx$ $= \frac{-1}{2e} - \frac{1}{e} + \left[-e^{-x^2} \right]_0^1$ $= \frac{-1}{2e} - \frac{1}{e} - \frac{1}{e} + 1$ $= \frac{-1}{2e} - \frac{2}{2e} - \frac{2}{2e} + 1$ $= 1 - \frac{5}{2e}$	2 2	2 – integration by parts to derive reduction formula 1 – first use of reduction formula 1 – simplifying to an answer
(b)	  (i) $x^2 + y^2 = 9$ $y = \sqrt{9 - x^2}$ $\therefore l = 2\sqrt{9 - x^2}$ $\sin 60^\circ = \frac{h}{l}$ $h = l \sin 60^\circ$ $h = \frac{\sqrt{3}}{2} l$ OR $A = \frac{1}{2} l^2 \sin 60^\circ$ $= \frac{\sqrt{3}}{4} l^2$	2	1 – expression for h

Question 15		2014	
	Solution	Marks	Allocation of marks
	$\therefore h = \sqrt{3} \sqrt{9 - x^2}$ $A(x) = \frac{1}{2}bh$ $= \frac{1}{2}(2\sqrt{9 - x^2})(\sqrt{3} \sqrt{9 - x^2})$ $= \sqrt{3}(9 - x^2)$ (ii) $V = \int_{-3}^3 \sqrt{3}(9 - x^2) dx$ $= \sqrt{3} \left[9x - \frac{x^3}{3} \right]_{-3}^3$ $= \sqrt{3}[(27 - 9) - (-27 + 9)]$ $= \sqrt{3}[18 + 18]$ $= 36\sqrt{3}$ OR $V = 2 \int_0^3 \sqrt{3}(9 - x^2) dx$ $= 2\sqrt{3} \left[9x - \frac{x^3}{3} \right]_0^3$ $= 2\sqrt{3} [(27 - 9) - 0]$ $= 2\sqrt{3} \times 18$ $= 36\sqrt{3}$	2	1 – expression for Area 1 - integral 1 - Answer

(c)	Let $f(x) = x^4 - 6x^3 + 9x^2 + 4x - 12$ $f'(x) = 4x^3 - 18x^2 + 18x + 4$ Double root when $f'(x) = f(x) = 0$ Test $x = \pm 1$ and $x = \pm 2$ (factors of 4) When $x = 2$, $f'(2) = 4(2^3) - 18(2^2) + 18(2) + 4$ $= 32 - 72 + 36 + 4 = 72 - 72 = 0$ $f(2) = (2^4) - 6(2^3) + 9(2^2) + 4(2) - 12$ $= 16 - 48 + 36 + 8 - 12 = 60 - 60 = 0$ $\therefore f'(2) = f(2) = 0$ $\therefore (x - 2)$ is a repeated factor. $\therefore \alpha = 2$ is a double root.	3	1 – using double root theorem and finding the derivative 1 – testing for roots of $f'(x)$ 1 – testing in $f(x)$ and stating the value of α
(d)	(i) $\arg z = \theta$ where $\tan \theta = \frac{y}{x}$ If $\arg(z) < \frac{\pi}{4}$ then $-\frac{\pi}{4} < \arg(z) < \frac{\pi}{4}$ 	2	1 – Graph 1 – showing main features

Question 16		2014	
	Solution	Marks	Allocation of marks
(a)	(i) $x = a \sec \theta$ $y = b \tan \theta$ $\frac{dx}{d\theta} = a \sec \theta \tan \theta$ $\frac{dy}{d\theta} = b \sec^2 \theta$ $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ $= \frac{b \sec^2 \theta}{a \sec \theta \tan \theta}$ $= \frac{b \sec \theta}{a \tan \theta}$ $y - y_1 = m(x - x_1)$ $y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$ $ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$ $-ay \tan \theta + bx \sec \theta = ab(\sec^2 \theta - \tan^2 \theta)$ Since $\sec^2 \theta - \tan^2 \theta = 1$ $bx \sec \theta - ay \tan \theta = ab$	2	1 – deriving gradient of tangent 1 – using equation to complete proof
	(ii) from (i) $m(\text{tangent}) = \frac{b \sec \theta}{a \tan \theta}$ $\therefore m(\text{normal}) = -\frac{a \tan \theta}{b \sec \theta}$ $y - y_1 = m(x - x_1)$ $y - b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$ $by \sec \theta - b^2 \tan \theta \sec \theta = -ax \tan \theta + a^2 \tan \theta \sec \theta$ By dividing by $\tan \theta \sec \theta$ $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$	2	1 – deriving gradient of normal 1 – equation of normal
	(iii) Tangent: $bx \sec \theta - ay \tan \theta = ab$ When $x = 0$ $y = \frac{-b}{\tan \theta} \therefore A\left(0, \frac{-b}{\tan \theta}\right)$ Normal: $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$ When $x = 0$ $y = \frac{(a^2 + b^2) \tan \theta}{b} \therefore B\left(0, \frac{(a^2 + b^2) \tan \theta}{b}\right)$	2	1 for A 1 for B

Question 16		2014	
	Solution	Marks	Allocation of marks
	(v) Focus of hyperbola = $S(ae, 0)$ If AB is diameter of circle then angle ASB must be right angled. $m(AS) = \frac{0 - \frac{-b}{\tan \theta}}{ae - 0}$ $= \frac{\frac{b}{\tan \theta}}{ae}$ $= \frac{b}{ae \tan \theta}$ $m(BS) = \frac{0 - \frac{(a^2 + b^2) \tan \theta}{b}}{ae - 0}$ $= - \frac{\frac{(a^2 + b^2) \tan \theta}{b}}{ae} = - \frac{(a^2 + b^2) \tan \theta}{abe}$ $m(AS) \times m(BS) = \frac{b}{ae \tan \theta} \times - \frac{(a^2 + b^2) \tan \theta}{abe}$ $= \frac{-(a^2 + b^2)}{a^2 e^2}$ Now $e^2 - 1 = \frac{b^2}{a^2}$ $e^2 = \frac{b^2}{a^2} + 1$ $= \frac{b^2 + a^2}{a^2}$ $\therefore m(AS) \times m(BS) = \frac{-(a^2 + b^2)}{a^2} \div \frac{b^2 + a^2}{a^2}$ $= \frac{-(a^2 + b^2)}{a^2} \times \frac{a^2}{b^2 + a^2}$ $= -1$ Therefore AB is diameter of circle passing through S, the foci of the hyperbola.	3	1 – gradients 1 - working 1 1 showing perpendicular

Question 16		2014	
	Solution	Marks	Allocation of marks
(b)	The letter 'S' occurs twice in CHRISTMAS Case 1: No S: Consider the letters CHRITMA Number of selections = 7C_5 and the possible arrangements of this selection is $5!$ Number with no S is ${}^7C_5 \times 5! = 2520$ Case 2: One 'S' so 4 from the remaining 7 letters = 7C_4 and the arrangements of this selection is $5!$. Number with 1 S is ${}^7C_4 \times 5! = 4200$ Case 3: Two 'S' so 3 from the remaining 7 letters = 7C_3 and the Arrangements of this selection is $\frac{5!}{2!}$. Number with 2 'S's is ${}^7C_3 \times \frac{5!}{2!} = 2100$ Total number of distinct arrangements $= 2520 + 4200 + 2100$ $= 8820$	2	1 working 1 correct answer
(c)	$\angle PAD = \angle AQC$ (alternate segment theorem) $\angle AQC = \angle CPD$ (alternate angles AQ parallel to PB) $\therefore \angle CPD = \angle PAD$ $\angle CDP = \angle PDA$ (common angle) $\therefore$ triangle CDP is similar to triangle PDA (equiangular)	2	1 – correct use of a circle geometry theorem 1 – correct proof
	(ii) triangle CDP is similar to triangle PDA $\therefore \frac{CD}{PD} = \frac{PD}{AD}$ (ratio of corresponding sides in similar triangles) $\therefore PD^2 = AD \times CD$ $DB^2 = DC \times DA$ (product of intercepts on secant equals square of tangent) i.e $DB^2 = PD^2$ $PD = DB$ $\therefore AD$ bisects PB	2	1 – correctly establishes result 1-correctly establishes result